

TIC 405216391

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 3.114526 d
Epoch (t0) 1492.2494 BTJD
Duration 0.1200 d
Depth 7.081 % (70,811 ppm)
SNR 233.5

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 8 (gate ≥ 3)
Robust depth ratio 0.96 (gate ≥ 0.50)
Robust depth sigma 111.3 (gate ≥ 7.0)
Odd/even z-score 2.48 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 7 light-curve dip in this sector: 139.4 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 137.43 (gate ≥ 5.0) Compactness 0.88 (gate ≥ 0.50)
Dimming centroid offset 0.20 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 405216381
0.35 px away, dTmag +0.05 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

